Subtracting (2) and (3),

$$\begin{array}{rl} x - y + z &= -2 \\ -5x - 4y + z &= -11 \quad (-) \\ \hline 6x + 3y &= 9 \end{array}$$

Dividing by 3 $\quad 2x + y = 3 \quad \ldots(5)$

Subtracting (4) and (5),

$$\begin{array}{rl} 2x + y &= 3 \\ 2x + y &= 3 \\ \hline 0 &= 0 \end{array}$$

Here we arrive at an identity 0=0.

Hence the system has an infinite number of solutions.

Example 3.6 Solve $3x + y - 3z = 1$; $-2x - y + 2z = 1$; $-x - y + z = 2$.

Solution $3x + y - 3z = 1 \quad \ldots (1) \qquad -2x - y + 2z = 1 \ \ldots(2) \qquad -x - y + z = 2 \ \ldots(3)$

Adding (1) and (2),

$$\begin{array}{rl} 3x + y - 3z &= 1 \\ -2x - y + 2z &= 1 \quad (+) \\ \hline x - z &= 2 \end{array} \qquad \ldots (4)$$

Adding (1) and (3),

$$\begin{array}{rl} 3x + y - 3z &= 1 \\ -x - y + z &= 2 \quad (+) \\ \hline 2x - 2z &= 3 \end{array} \qquad \ldots (5)$$

Now, (5) $-2\times$(4) we get,

$$\begin{array}{rl} 2x - 2z &= 3 \\ 2x - 2z &= 4 \quad (-) \\ \hline 0 &= -1 \end{array}$$

Here we arrive at a contradiction as $0 \neq -1$.

This means that the system is inconsistent and has no solution.

Example 3.7 Solve $\frac{x}{2} - 1 = \frac{y}{6} + 1 = \frac{z}{7} + 2$; $\frac{y}{3} + \frac{z}{2} = 13$

Solution Considering, $\frac{x}{2} - 1 = \frac{y}{6} + 1$

$$\frac{x}{2} - \frac{y}{6} = 1 + 1 \Rightarrow \frac{6x - 2y}{12} = 2 \text{ we get, } 3x - y = 12 \qquad \ldots (1)$$

Considering, $\frac{x}{2} - 1 = \frac{z}{7} + 2$

$$\frac{x}{2} - \frac{z}{7} = 1 + 2 \Rightarrow \frac{7x - 2z}{14} = 3 \text{ we get, } 7x - 2z = 42 \qquad \ldots (2)$$

Also, from $\frac{y}{3} + \frac{z}{2} = 13 \Rightarrow \frac{2y + 3z}{6} = 13$ we get, $2y + 3z = 78 \quad \ldots (3)$

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